MIS
American International University-Bangladesh

AIUB FBA FACULTY OF BUSINESS ADMINISTRATION

MANAGEMENT INFORMATION SYSTEMS

Managing the Digital Firm

Laudon and Laudon, 18th Edition

Chapter 2

GLOBAL E-BUSINESS AND COLLABORATION

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Business Processes and Information Systems

- **Business processes:**
  - Workflows of material, information, knowledge
  - Sets of activities, steps
  - May be tied to functional area or be cross-functional
- **Businesses: Can be seen as collection of business processes**
- **Business processes may be assets or liabilities**





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Business Processes and Information Systems

- **Examples of functional business processes**
  - **Manufacturing and production**
    - Assembling the product; checking for quality of the product; producing bills of materials
  - **Sales and marketing**
    - Identifying customers; making customers aware of the product; selling the product
  - **Finance and accounting**
    - Paying creditors; creating financial statements; managing cash accounts
  - **Human resources**
    - Hiring employees; evaluating employees' job performance; enrolling employees in benefits plans

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Business Processes and Information Systems

Table 2.1 Examples of Functional Business Processes

Functional Area	Business Process
Manufacturing and production	Assembling the product Checking for quality of the product Producing bills of materials
Sales and marketing	Identifying customers Making customers aware of the product Selling the product
Finance and accounting	Paying creditors Creating financial statements Managing cash accounts
Human resources	Hiring employees Evaluating employees' job performance Enrolling employees in benefits plans

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Business Processes and Information Systems

The Order Fulfillment Process

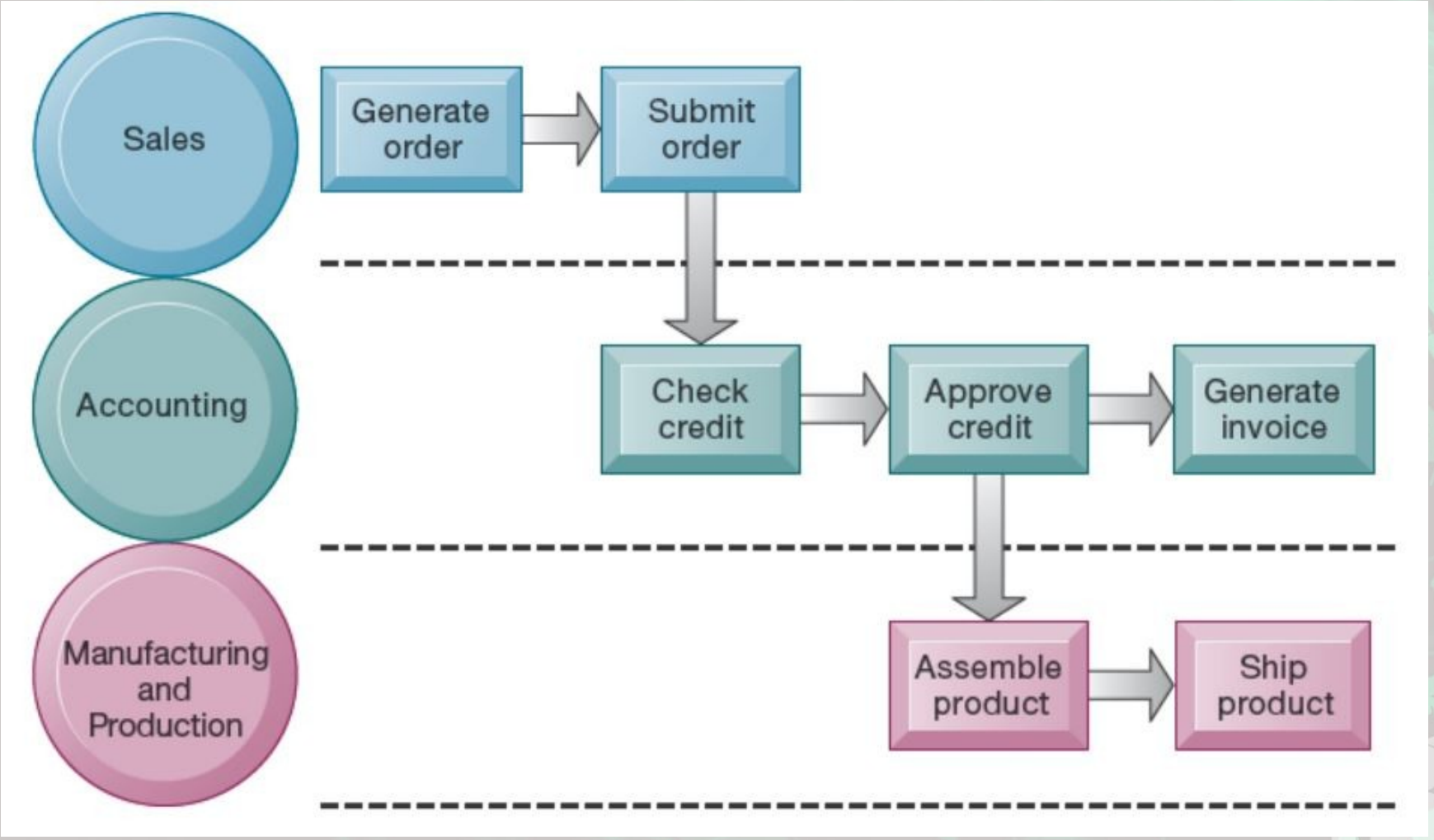


Figure 2.1

Fulfilling a customer order involves a complex set of steps that requires the close coordination of the sales, accounting, and manufacturing functions.

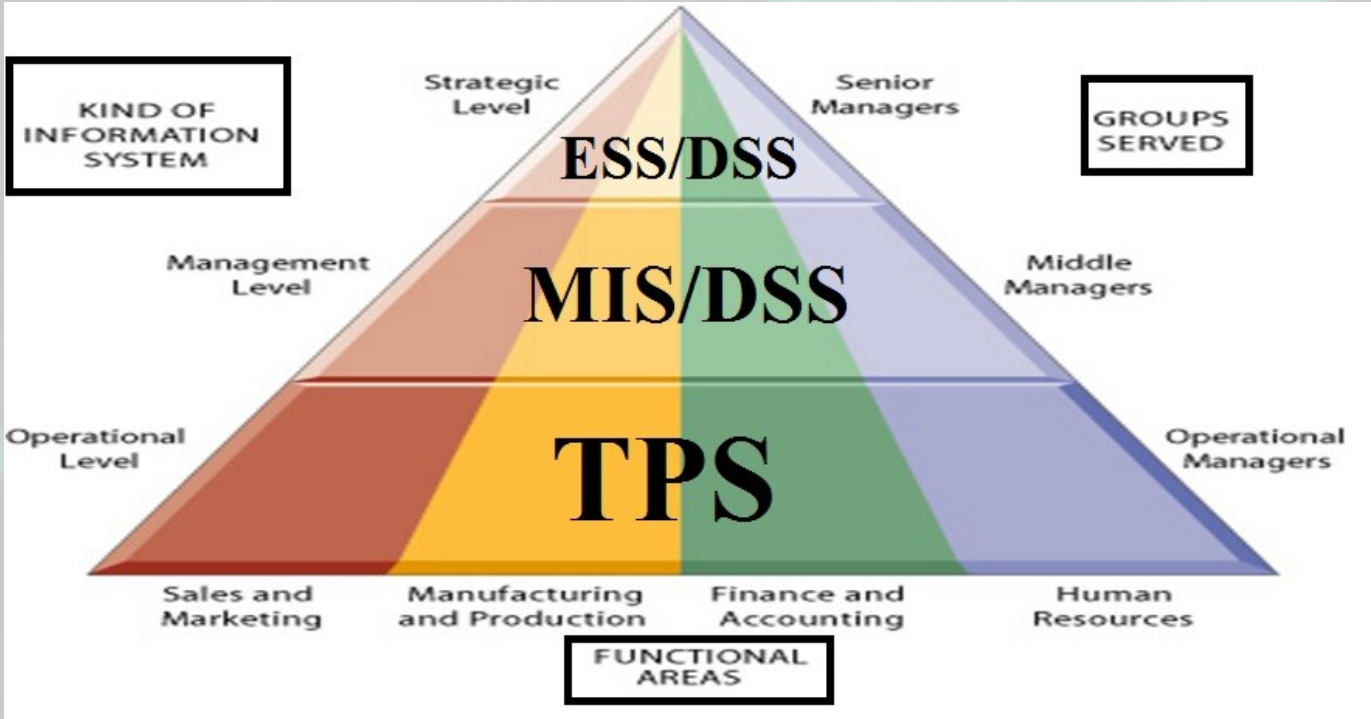
Business Processes and Information Systems

- **Information technology enhances business processes in two main ways:**
  1. Increasing efficiency of existing processes
     - Automating steps that were manual
  2. Enabling entirely new processes that are capable of transforming the businesses
     - Change flow of information
     - Replace sequential steps with parallel steps
     - Eliminate delays in decision making

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Types of Information Systems

- Transaction Processing Systems (TPS)
- Management Information Systems (MIS)
- Decision Support Systems (DSS)
- Executive Support Systems (ESS)



Types of Information Systems

- **Transaction Processing Systems (TPS)**
  - Perform and record daily routine transactions necessary to conduct business
    - Examples: sales order entry, payroll, shipping
  - Allow managers to monitor status of operations and relations with external environment
  - Serve operational levels
  - Serve predefined, structured goals and decision making

Types of Information Systems

A Payroll TPS

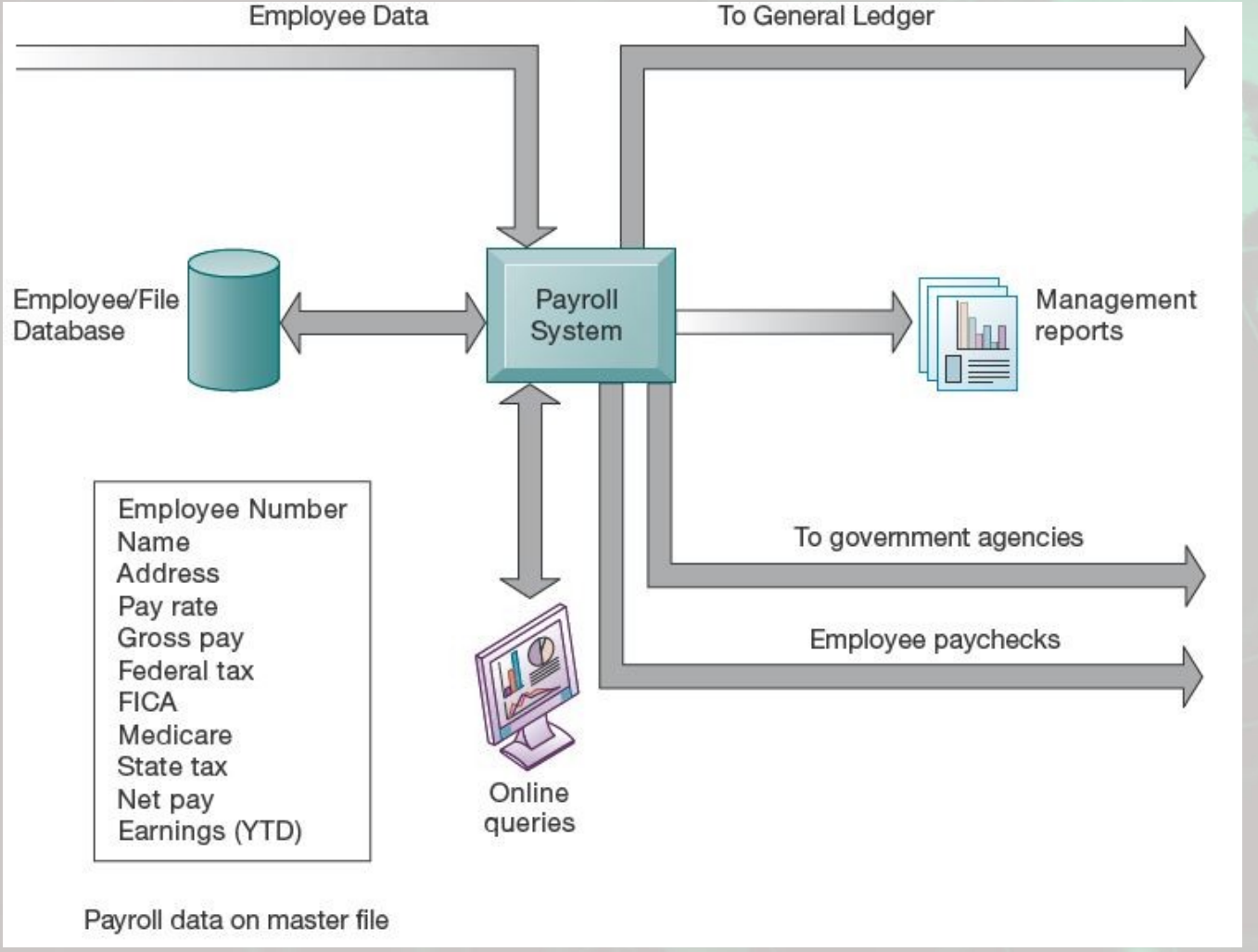


Figure 2.2

A TPS for payroll processing captures employee payment transaction data (such as a timecard). System outputs include online and hard-copy reports for management and employee paychecks.

Types of Information Systems

- **Management Information Systems (MIS)**
  - Serve middle management
  - Provide reports on firm's current performance, based on data from TPS
  - Provide answers to routine questions with predefined procedure for answering them
  - Typically have little analytic capability

Types of Information Systems

How Management Information Systems Obtain Their Data from the Organization's TPS

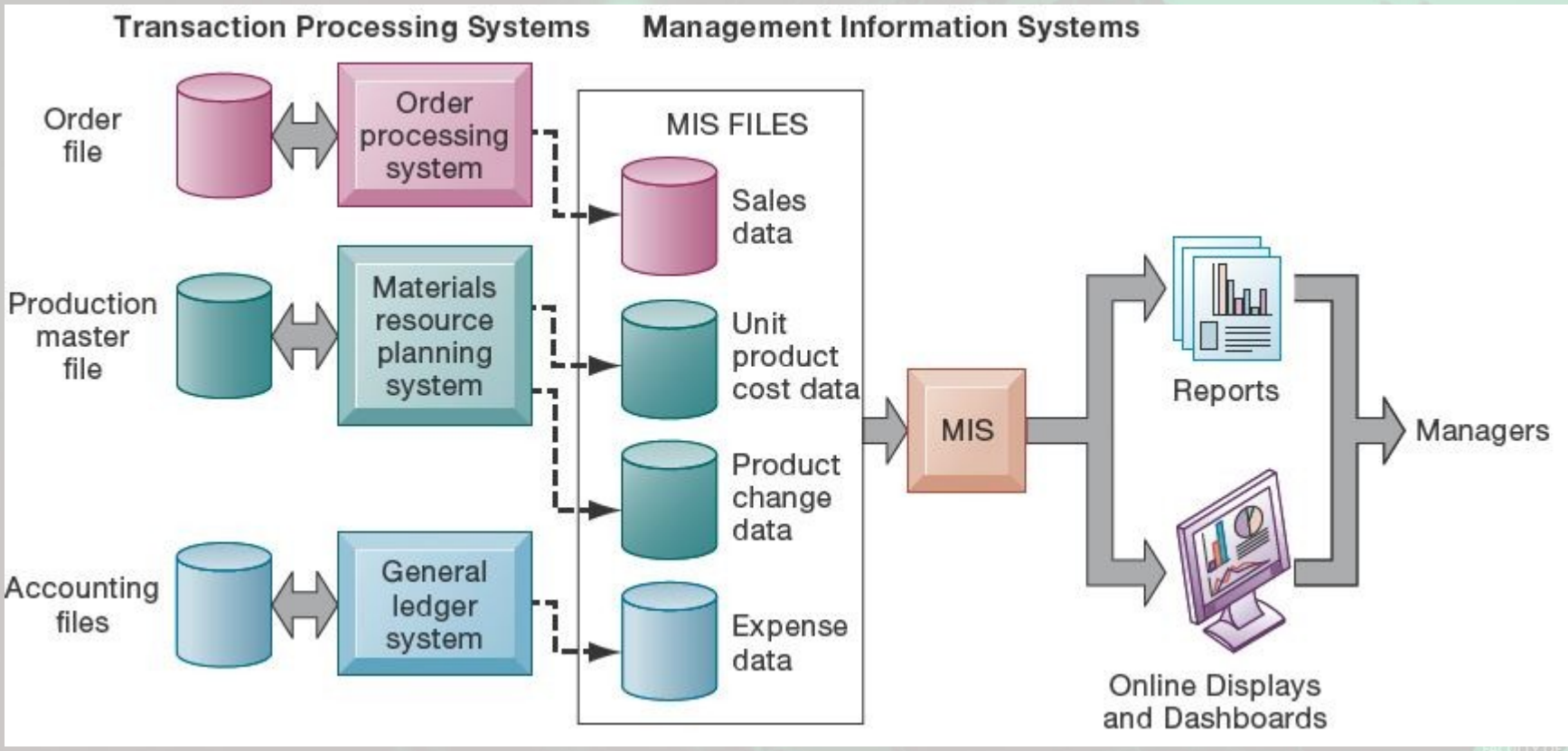


Figure 2.3

In the system illustrated by this diagram, three TPS supply summarized transaction data to the MIS reporting system at the end of the time period. Managers gain access to the organizational data through the MIS, which provides them with the appropriate reports.

Types of Information Systems

Sample MIS Report

Consolidated Consumer Products Corporation Sales by Product and Sales Region: 2020

PRODUCT CODE	PRODUCT DESCRIPTION	SALES REGION	ACTUAL SALES	PLANNED	ACTUAL versus PLANNED
4469	Carpet Cleaner	Northeast	4,066,700	4,800,000	0.85
		South	3,778,112	3,750,000	1.01
		Midwest	4,867,001	4,600,000	1.06
		West	4,003,440	4,400,000	0.91
	TOTAL		16,715,253	17,550,000	0.95
5674	Room Freshener	Northeast	3,676,700	3,900,000	0.94
		South	5,608,112	4,700,000	1.19
		Midwest	4,711,001	4,200,000	1.12
		West	4,563,440	4,900,000	0.93
	TOTAL		18,559,253	17,700,000	1.05

Figure 2.4

This report, showing summarized annual sales data, was produced by the MIS in Figure 2-3.

Types of Information Systems

- **Decision Support Systems (DSS)**
  - Serve middle management
  - Support non-routine decision making
    - Example: Voyage-estimating DSS used by a global shipping company to develop bids on shipping contracts (Figure 2.5)
    - Calculates financial voyage details (ship/time, fuel, port costs, freight rates) and technical details (capacity, speed, distances) to maximize profit
  - Often use external information as well from TPS and MIS

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Types of Information Systems

- **Business Intelligence (BI)**
  - Class of software applications
  - Analyze current and historical data to find patterns and trends and aid decision-making
  - Used in systems that support middle and senior management
    - Data-driven DSS
    - Executive Support Systems (ESS)

Types of Information Systems

- **Executive Support Systems (ESS)**
  - Support senior management
  - Address non-routine decisions
    - Requiring judgment, evaluation, and insight
  - Incorporate data about external events (e.g. new tax laws or competitors) as well as summarized information from internal MIS and DSS
  - Example: Digital dashboard with real-time view of firm's financial performance: working capital, accounts receivable, accounts payable, cash flow, and inventory

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Types of Information Systems

- **Systems from a constituency perspective**
  - **TPS:**
    - Supporting operational level employees
  - **MIS and DSS:**
    - Supporting middle managers
  - **ESS:**
    - Supporting executives

Types of Information Systems

- **Relationship of systems to one another**
  - **TPS:** Major source of data for other systems
  - **ESS:** Recipient of data from lower-level systems
  - Data may be exchanged between systems
  - In reality, most businesses' systems are only loosely integrated (but they are getting better!)

Types of Information Systems

- **Enterprise applications**
  - Systems for linking the enterprise
  - Span functional areas
  - Execute business processes across firm
  - Include all levels of management
  - Four major applications:
    - **Enterprise Systems (ES)**
    - **Supply Chain Management Systems (SCMS)**
    - **Customer Relationship Management Systems (CRMS)**
    - **Knowledge Management Systems (KMS)**

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Types of Information Systems

Enterprise Application Architecture

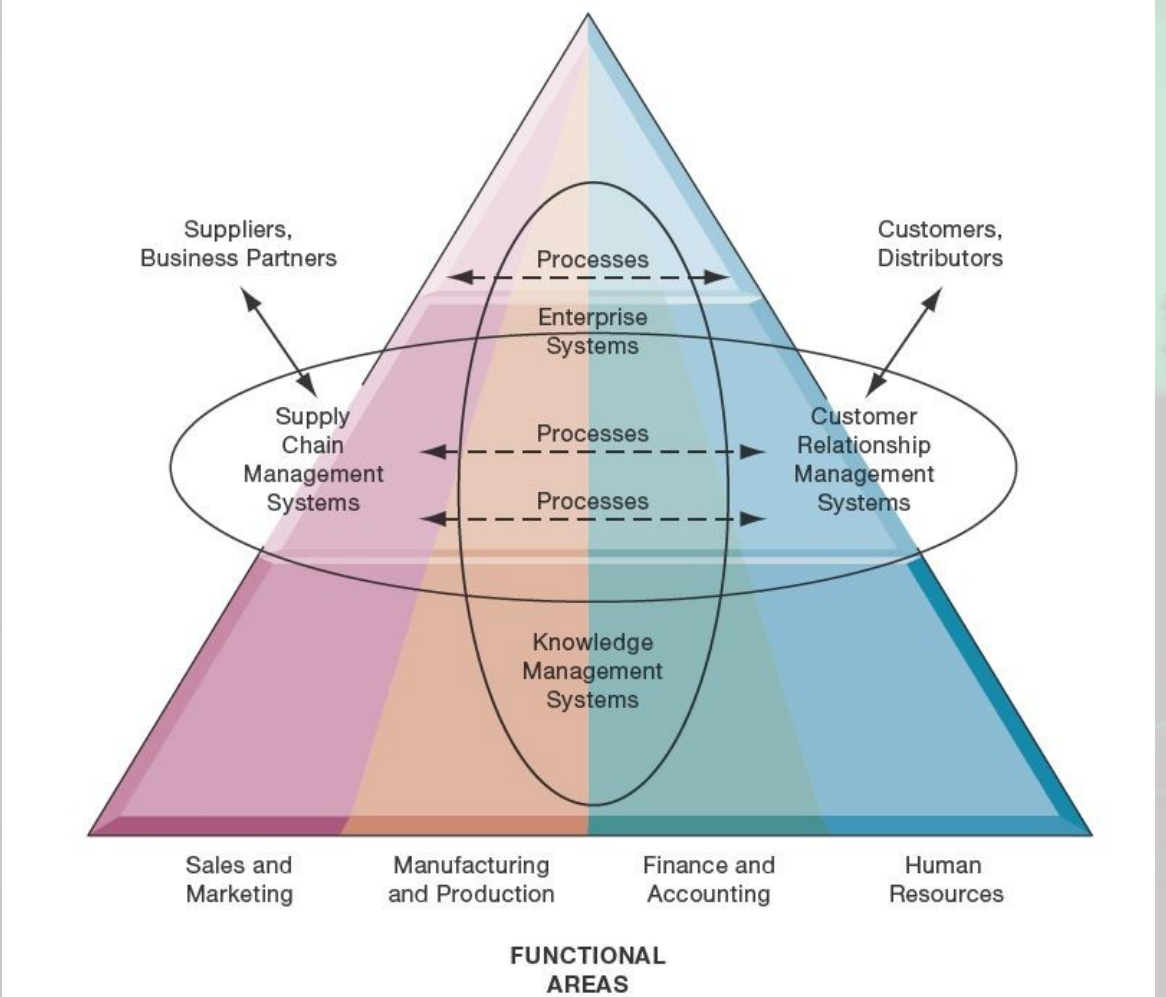


Figure 2.5

Enterprise applications automate processes that span multiple business functions and organizational levels and may extend outside the organization.

Types of Information Systems

- **Enterprise Systems**
  - Collect data from different firm functions and stores data in single central data repository
  - Resolve problem of fragmented, redundant data sets and systems
  - Enable:
    - Coordination of daily activities
    - Efficient response to customer orders (production, inventory)
    - Provide valuable information for improving management decision making

Usefulness of ERP for Business Analysts (Supplementary)

- **Data Integration and Centralization:** ERP integrates business functions into a unified platform, centralizing data from finance, sales, and inventory to provide comprehensive, integrated insights that enhance decision-making.
- **Enhanced Decision-Making:** Real-time access to and interpretation of data helps identify areas for cost reduction and revenue growth, supporting strategic initiatives.
- **Improved Forecasting:** Predictive analytics features let users forecast future trends from historical data, enabling proactive strategies and improved operational efficiency.

Usefulness of ERP for Business Analysts (Supplementary, cont.)

- **Operational Efficiency and Process Optimization:** ERP streamlines business processes by automating routine tasks and identifying bottlenecks, helping analysts recommend improvements that enhance productivity and reduce operational costs.
- **Real-World Application of Business Intelligence:** BI tools complement ERP systems and provide deeper insights into organizational performance, enabling dashboards and reports that support strategic planning and operational adjustments.
- **Real-Time Decision Making:** ERP provides real-time information for timely decisions, enhancing the ability to respond quickly to changing market conditions and operational challenges.

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Types of Information Systems

- **Supply Chain Management Systems (SCMS)**
  - Manage firm's relationships with suppliers
  - Share information about
    - Orders, production, inventory levels, delivery of products and services
  - Goal:
    - Right amount of products to destination with least amount of time and lowest cost

Types of Information Systems

- **Customer Relationship Management Systems (CRMS):**
  - Provide information to coordinate all the business processes that deal with customers in sales, marketing, and service to optimize revenue, customer satisfaction, and customer retention
  - Integrate firm's customer-related processes and consolidate customer information from multiple communication channels

Types of Information Systems

- **Knowledge Management Systems (KMS):**
  - Support processes for acquiring, creating, storing, distributing, applying, integrating knowledge
    - How to create, produce, distribute products and services
  - Collect internal knowledge and experience within firm and make it available to employees
  - Link to external sources of knowledge

Types of Information Systems

- **Alternative tools that increase integration and expedite the flow of information**
  - Intranets:
    - Internal company Web sites accessible only by employees
  - Extranets:
    - Company Web sites accessible externally only to vendors and suppliers
    - Often used to coordinate supply chain

Types of Information Systems

- **E-business**
  - Use of digital technology and Internet to drive major business processes
- **E-commerce**
  - Subset of e-business
  - Buying and selling goods and services through Internet
- **E-government:**
  - Using Internet technology to deliver information and services to citizens, employees, and businesses

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Systems for Collaboration and Teamwork

- **Technology for collaboration and teamwork (cont.)**
  - Social Networking
  - Wikis
  - Virtual Worlds
  - Internet-Based Collaboration Environments
    - Virtual meeting systems (telepresence)
    - Google Apps/Google sites
    - Microsoft SharePoint
    - Lotus Notes

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Systems for Collaboration and Teamwork

Table 2.2 Applications of Social Business

Application	Description
Social networks	Connect through personal and business profiles
Crowdsourcing	Harness collective knowledge to generate new ideas and solutions
Shared workspaces	Coordinate projects and tasks; co-create content
Blogs and wikis	Publish and rapidly access knowledge; discuss opinions and experiences
Social commerce	Share opinions about purchasing on social platforms
File sharing	Upload, share, and comment on photos, videos, audio, text documents
Social marketing	Use social media to interact with customers; derive customer insights
Communities	Discuss topics in open forums; share expertise

Systems for Collaboration and Teamwork

Table 2.3 Business Benefits of Collaboration and Social Business

Benefit	Rationale
Productivity	People interacting and working together can capture expert knowledge and solve problems more rapidly than the same number working in isolation. There will be fewer errors.
Quality	People working collaboratively can communicate errors and corrective actions faster than if they work in isolation, reducing time delays in design and production.
Innovation	Collaborative groups can generate more innovative ideas for products, services, and administration, gaining advantages from diversity and the "wisdom of crowds."
Customer service	Teams using collaboration and social business tools can solve customer complaints and issues faster and more effectively than individuals working alone.
Financial performance	As a result of all of the above, collaborative firms have the potential for superior sales, sales growth, and profitability.

Systems for Collaboration and Teamwork

Table 2.4 Examples of Human-AI Collaboration Applications

Task	Example
Medical diagnoses	AI algorithms analyze medical images (X-rays, MRIs, CT scans, ECGs) to detect abnormalities; radiologists collaborate with AI to validate findings and make final diagnoses.
Aggregating and synthesizing reviews	CarMax used the GPT-3.5 language model to synthesize 5,000 online customer reviews; after human review, the descriptions were posted on its website.
Selecting financial instruments	Traders at JP Morgan Chase work with a ChatGPT-like model that analyzed 25 years of Federal Reserve speeches to decipher trading signals.
Customized manufacturing	BionicWorkplace's intelligent workstation uses a bionic arm, sensors, and cameras to derive the optimal human-robot work sequence and continuously learns from each action.

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Systems for Collaboration and Teamwork

Table 2.5 Enterprise Social Networking Software Capabilities

Capability	Description
Profiles	Set up member profiles describing background, interests, and work-related associations and expertise (skills, projects, teams).
Content sharing	Share, store, and manage content including documents, presentations, images, and videos.
Feeds and notifications	Real-time information streams, status updates, and announcements from designated individuals and groups.
Groups and team workspaces	Establish private and public groups to share information and collaborate on documents; archive conversations to preserve team knowledge.
Tagging and social bookmarking	Indicate preferences for content (like the Facebook Like button); add keywords to identify content of interest.
Permissions and privacy	Ensure private information stays within the right circles by setting who in the company can see what information.

Systems for Collaboration and Teamwork

- **Two dimensions of collaboration technologies**
  - Space (or location) – remote or collocated
  - Time – synchronous or asynchronous
- **Six steps in evaluating software tools**
  1. What are your firm's collaboration challenges?
  2. What kinds of solutions are available?
  3. Analyze available products' cost and benefits
  4. Evaluate security risks
  5. Consult users for implementation and training issues
  6. Evaluate product vendors

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The Information Systems Function in Business

- **Information Systems Department:**
  - Formal organizational unit responsible for information technology services
  - Often headed by Chief Information Officer (CIO)
    - Other senior positions include Chief Security Officer (CSO), Chief Knowledge Officer (CKO), Chief Privacy Officer (CPO)
  - Programmers
  - Systems analysts
  - Information systems managers

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The Information Systems Function in Business

- **End Users**
  - Representatives of other departments for whom applications are developed
  - Increasing role in system design, development
- **IT Governance:**
  - Strategies and policies for using IT in the organization
  - Decision rights
  - Accountability
  - Organization of information systems function
    - Centralized, decentralized, etc.

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Thank you

Chapter 2

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